

SMC Reset Unit — Test Plan

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Verification plan for the SMC Reset Unit SystemC/TLM-2.0 LT model. Two self-checking benches (`test/reset_unit_tb.cpp`, `test/reset_unit_neg_tb.cpp`) print `ALL TESTS PASSED` on success and exit non-zero on any failure; both are registered with CTest.

1. Strategy

- **Functional bench** (`reset_unit_tb`) wires the DUT into a full SystemC elaboration (all ports bound to signals, a TLM initiator on the register socket) and drives stimulus from an `SC_THREAD`, checking register read-backs, derived reset outputs, the per-subsystem bundles, and the FLR pulse timing.
 - **Negative bench** (`reset_unit_neg_tb`) probes constructor guard rails (which `SC_REPORT_FATAL/throw`) and the TLM error taxonomy. It runs entirely during elaboration (direct `b_transport` calls, no `sc_start`) so the constructor-throw probes cannot perturb a running kernel — the same pattern the sister-IP negative benches use.
 - **AddressSanitizer** is run via `run_tests.sh --asan` to confirm no memory errors.
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2. Functional bench coverage (reset_unit_tb)

#	Test	Checks
1	Out-of-reset derivation	All derived resets de-asserted (high) once pins released.
2	RW register access	Reset defaults (SS_WARM_RESET_N = 0xFFFFFFFF); write/read of SS_WARM_RESET_N, SS_CONFIG_HOLD, SS_DEBUG_HOLD.
3	Lock semantics	SS_CONFIG_LOCK woset (sticky, accumulating); locked SS_CONFIG bits write-protected.
4	Per-subsystem bundle	SS_COLD_RESET_N / SS_WARM_RESET_N / SS_SRAM_HOLD / SS_FORCE_T0_REF_CLK bits map to ss_reset_ctrl_o[i] and dbg_ss_reset_ctrl(i).
5	ss_config_o	SS_CONFIG value reflected on the output port.
6	Straps	STRAPS_LO / STRAPS_HI reflect captured_straps_i.
7	Cold reset	Asserts cold/primary/core; clears all registers; releases to defaults.
8	WDT 2nd timeout	Forces core + WDT reset, leaves primary up.
9	Fuse reset	Gates core reset only.
10	JTAG overrides	Forced cold reset; per-subsystem warm-reset override; clear restores.
11	Isolate logic	ISOLATE_REQ_REG + pin-enable (ISOLATE_REQ_PINEN_REG & pin); skip_mem_repair_o; ISOLATE_REQ_VIS.
12	FLR cool-reset	FLR rising edge sets ISOLATE_REQ_SMC_REG, asserts smc-enabled isolate, skip_mem_repair_o; rst_cool_no pulses low after the delay and releases after the hold; SW write clears the latch.
13	Reset domains	Cool (from-pin) reset clears primary-domain regs but retains FLR/isolate (cold-domain) regs.
14	SYNC_REG	Drives sync_irq_o.
15	transport_dbg	Back-door write/read round-trip.
16	CCI	Param discovery; mutate access_delay_ns; num_subsystems immutability.
17	dump_state	Snapshot contains expected fields.

3. Negative bench coverage (reset_unit_neg_tb)

#	Test	Checks
1	Constructor guard rails	num_subsystems = 0 and = 33, ref_clk_period_ns < 0, access_delay_ns < 0 each raise SC_REPORT_FATAL.
2	TLM error taxonomy	Non-word width, misaligned, streaming-width mismatch, byte-enable, out-of-window, bad command → the canonical responses (spec §10).
3	RAZ/WI hole	A read/write of an unmapped offset returns TLM_OK_RESPONSE with zero data.
4	transport_dbg	Malformed accesses return 0; a valid write/read round-trips.

4. Running

```
# C++20 toolchain (repo standard)
SYSTEMC_HOME=/Users/pdroy/local/systemc-3.0.2-cxx20 \
CCI_HOME=/Users/pdroy/local/cci-cxx20 \
./run_tests.sh --clean          # build + run both benches

./run_tests.sh --ctest         # via CTest (2/2)
./run_tests.sh --asan          # AddressSanitizer
```

5. Results (reference run)

Metric	Result
reset_unit_tb	ALL TESTS PASSED (17 groups)
reset_unit_neg_tb	ALL TESTS PASSED
CTest	2/2 passed
AddressSanitizer	0 errors

6. Gaps / future tests

- Cycle-accurate de-glitch / extend / stretch timing (currently abstracted).
- Register write-gating while a reset is asserted (currently permitted).
- Multi-subsystem FLR / isolate sweeps and randomised register soak.